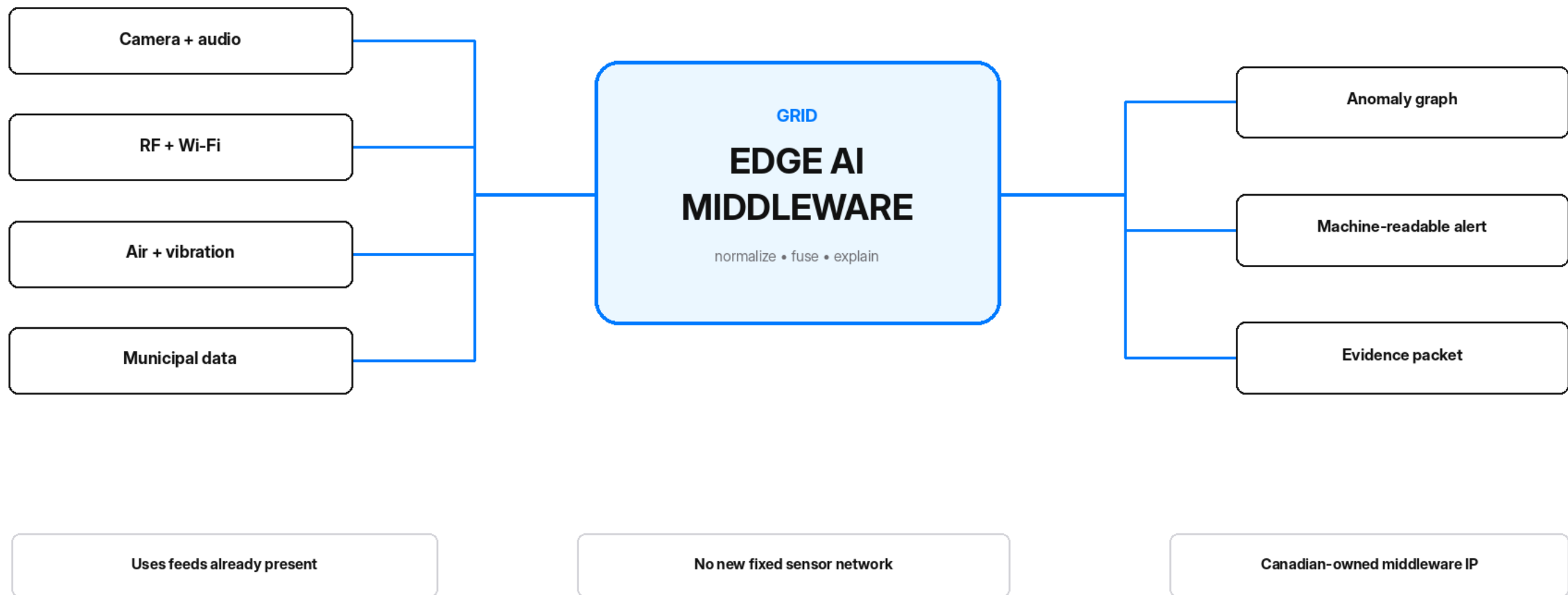


Grid

AI middleware that turns existing urban infrastructure into passive, explainable, real-time insight



Built for the evaluation rubric

02

The deck mirrors the mandatory and point-rated criteria.

Criterion	Evaluator signal	Points
MC-1	TRL evidence	Pass
MC-2	Challenge alignment	Pass
PRC-1	Scientific / technical merit	10
PRC-2	Novelty	20
PRC-3	Impact	20
PRC-4	Feasibility	20
PRC-5	GBA Plus	5
PRC-6	Desired outcomes	15
PRC-7	Cost alignment	10

70+ minimum responsive score

100 available PRC points

12 months Component 1b period

\$1.48M proposed request

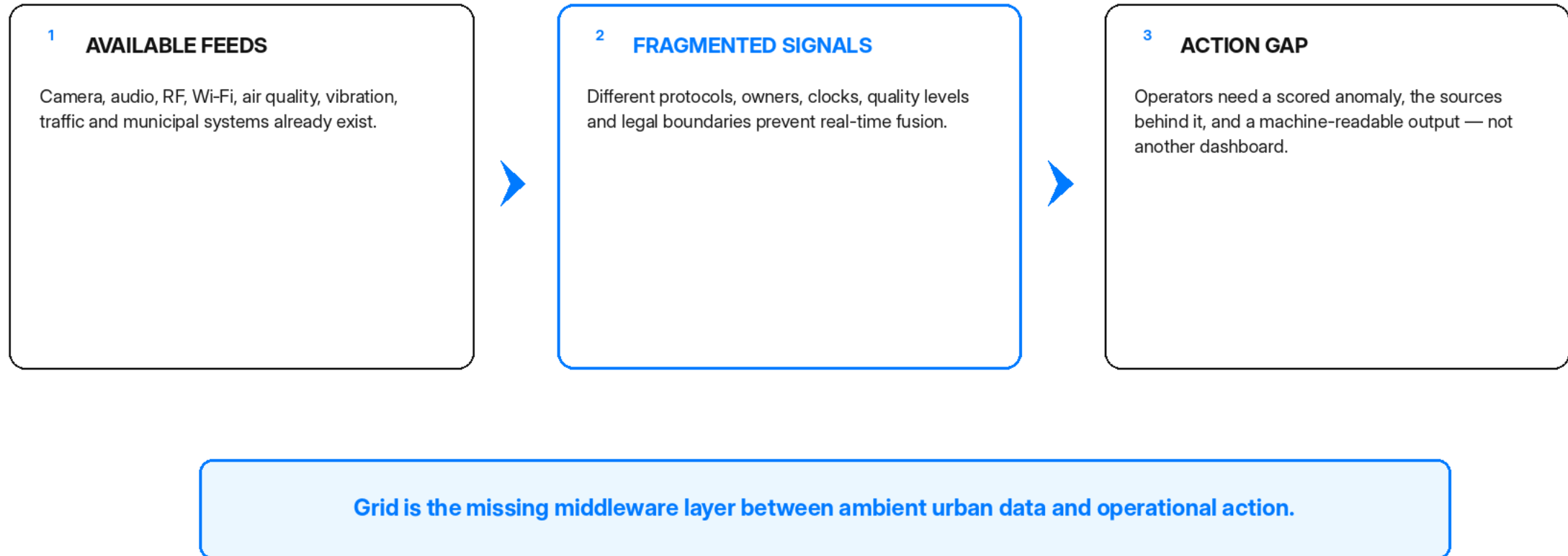
Design principle: make scoring effortless.

Source basis: challenge page, tender notice, and CFP criteria.

The city is instrumented — but not operationally aware

03

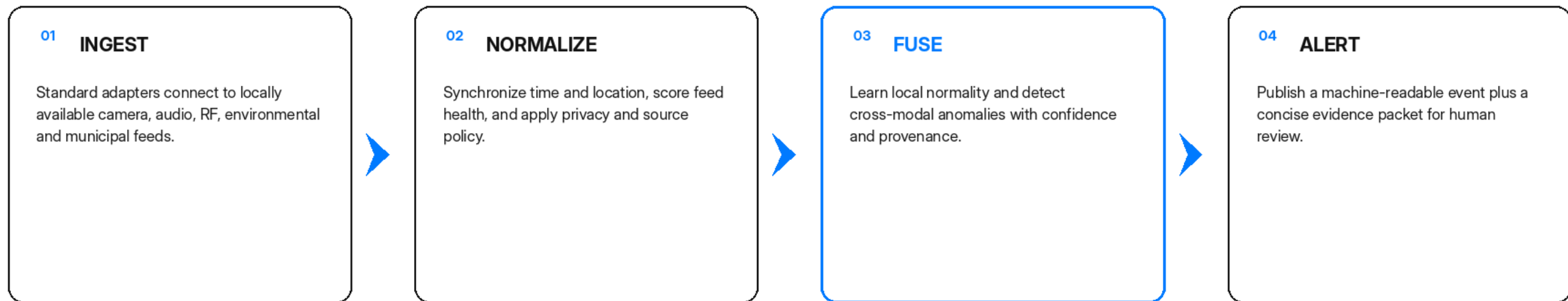
The problem is fragmentation, not a lack of data.



Grid: an edge-deployable AI middleware layer

04

A four-stage pipeline with one output contract.



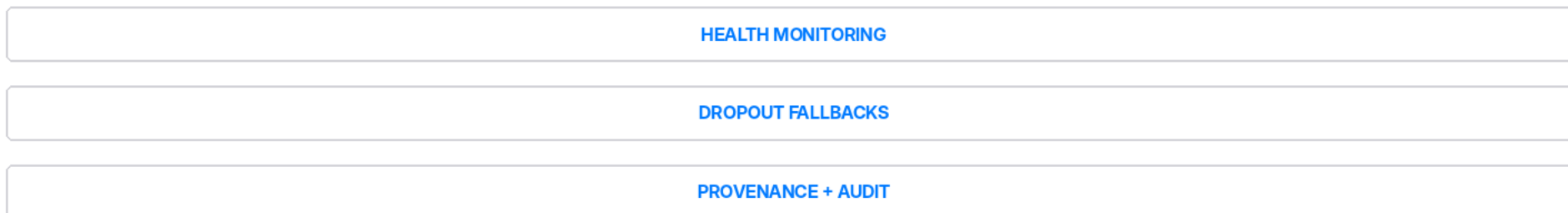
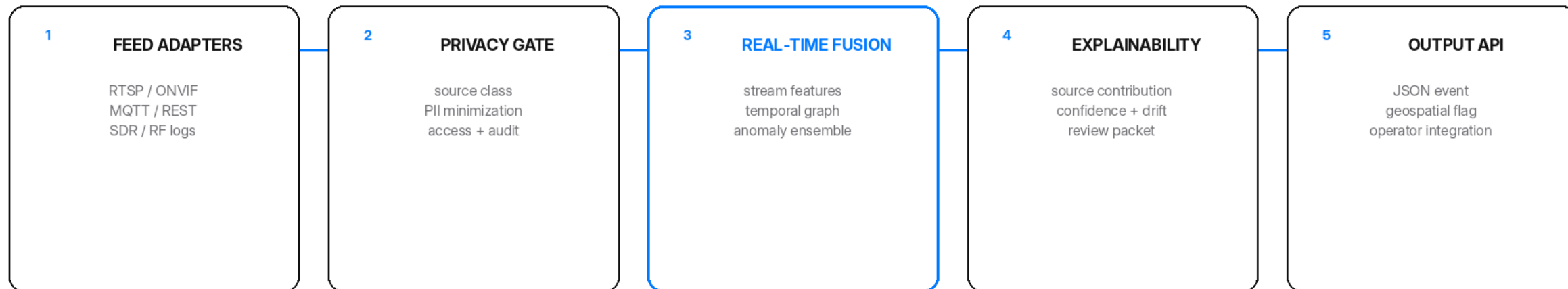
Zero new fixed hardware required. Grid adapts to the feeds available on arrival.

Essential outcomes: heterogeneous ingestion • real-time analysis • machine-readable output • legal/privacy compliance

Architecture: resilient by design

05

Five modules share one baseline, one audit trail, and one output schema.



Every module is independently testable and can run edge-only or hybrid.

Three modules prove operational impact

One middleware layer; three DND/CAF-relevant demonstrations.

01 COUNTER-UAS

Fuse camera motion, passive RF and acoustic cues to detect, classify and track unauthorized drones.

OUTPUT

track • confidence • source evidence

02 CRITICAL INFRASTRUCTURE

Correlate vibration, access, network, camera and environmental anomalies around bridges, power, water and transport assets.

OUTPUT

incident score • location • source evidence

03 ENVIRONMENTAL WARNING

Use air, weather and secondary signals to surface smoke, plume, contamination or abnormal operating conditions.

OUTPUT

event score • trend • source evidence

The same event schema lets operator tools consume all three use cases.

Novelty: a sovereign anomaly-fusion engine

07

Differentiated by data scarcity, explainability, resilience, and privacy.

01

SELF-SUPERVISED NORMALITY

Learns what is normal by place and time without labelled threat datasets.

02

SENSOR-DROPOUT RESILIENCE

Continues scoring when feeds degrade, disappear, or change protocol.

03

SYNTHETIC + SURROGATE DATA

Builds representative tests because DND/CAF will not provide training data.

04

EVIDENCE-FIRST OUTPUT

Every alert carries confidence, source contribution, provenance and drift status.

WHY THIS SCORES

New knowledge

Improved capability

Reusable Canadian IP

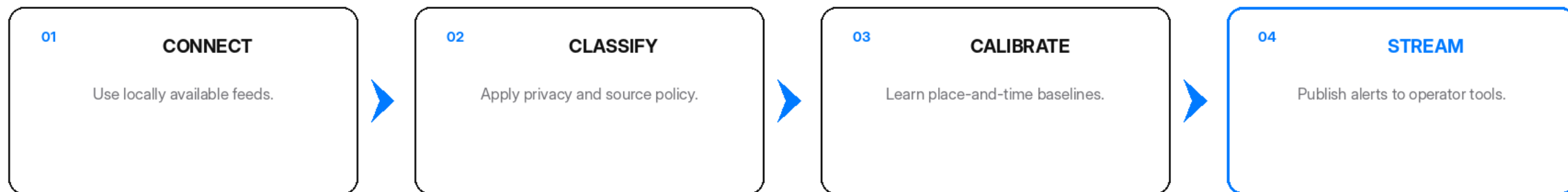
Auditable outputs

PRC-2 target: 20 / 20

Rapid deployment: operational within hours

08

A consistent four-step sequence in every new urban environment.



Fast setup • standard protocols • adaptive ingestion • explainable evidence • scalable architecture

Evidence plan: measurable, auditable tests

The prototype builds its own representative corpus and validates every claim.

01 DATA CORPUS

Public urban video, audio, RF and environmental datasets.

Synthetic digital-twin events for rare or unsafe anomalies.

Controlled municipal and commercial surrogate feeds.

02 TEST HARNESS

Milestone entry and exit criteria.

Red-team anomaly injection.

Feed dropout and drift trials.

Model cards, data cards and complete audit logs.

03 KPI TARGETS

< 5 s edge alert latency

≥ 2 heterogeneous sources per event

90% explainability packets complete

30-day drift and dropout resilience trial

No dependency on DND/CAF training, fine-tuning, or validation data.

Privacy, legal, and GBA Plus are system requirements

10

Controls are aligned across source, processing, and output.

01 SOURCE CONTROLS

Classify source and legal basis.

Minimize personally identifiable information.

Apply role-based access and retention policy.

02 PROCESSING CONTROLS

Encrypt in transit and at rest.

Prefer edge processing to reduce movement.

Maintain event-level provenance and audit.

03 OUTPUT CONTROLS

Require human review before consequential action.

Expose confidence and coverage gaps.

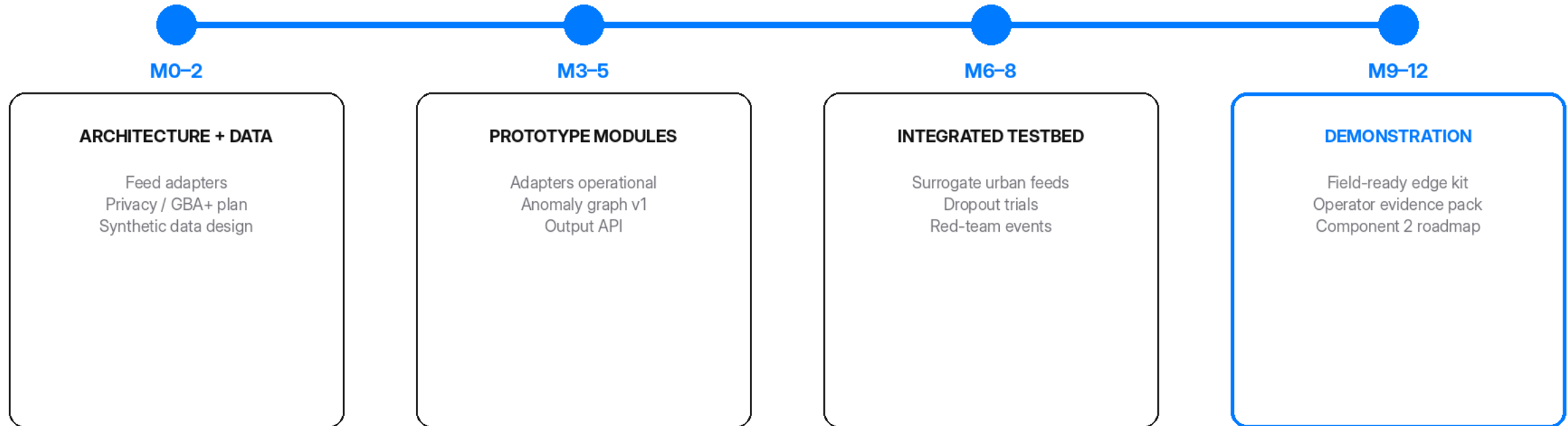
Test false positives and differential impacts.

Privacy by design • GBA Plus checkpoints • operator accountability • auditable evidence

Component 1b work plan: four gated phases

11

A 12-month path from TRL 4/5 to a TRL 6 demonstration package.



Risk controls: synthetic/surrogate data • source-classification gates • adaptive adapters • continuous drift checks • human review

The ask: fund the reusable urban intelligence layer

12

Grid is Canadian-owned, challenge-aligned, and ready for Component 1b.

\$1.48M

Component 1b request

12 MONTHS

performance period

TRL 6

demonstration exit

BUDGET LOGIC

Labour maps directly to four gated milestones.

Equipment is limited to portable edge compute, radios and test instrumentation.

Every deliverable advances technical evidence, not overhead.

WHY GRID NOW

Meets all essential outcomes.

Addresses every desired outcome.

Creates sovereign Canadian IP without relying on DND/CAF training data.

Establishes a credible Component 2 path.

Decision: invite Grid into Component 1b.

Appendix: source alignment

Reviewer-facing compliance trail.

Challenge objective	Repurpose existing urban infrastructure as a passive sensing network for scalable, real-time situational awareness and anomaly detection.
Essential outcomes	Ingest heterogeneous sources; analyze in real time; produce machine-readable alerts; comply with legal, privacy, and data-protection requirements.
Desired outcomes	Rapid deployability; adaptive ingestion; explainable outputs; scalable architecture compatible with international standards.
Evaluation criteria	Mandatory TRL and challenge alignment; 100 point-rated points; 70-point minimum; novelty, impact, and feasibility carry the largest weights.
Funding fit	Component 1b targets TRL 4/5, up to \$1.5M and 12 months, with a Component 2 build and validation path.

Verify final solicitation requirements and submission instructions before filing.

Grid operating dashboard

14

Real-time anomaly awareness from city infrastructure without a new fixed sensor network.

